Blockchain-Enabled Decentralized Services and Networks: Assessing Roles and Impacts

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***Abstract*—The rapid evolution of blockchain has established it as a critical enabler for decentralized zero-trust services and networks. Without relying on traditional trust mechanisms such as pre-established mutual trust or central authentication, blockchain facilitates trust-free services via smart contract. Smart contracts offer verifiable software trust for various blockchain-enabled services (BESs) while protecting participants' interests. However, the impact of blockchain on BES remains underexplored and unclear. In this work, we consider a general BES framework suitable for diverse decentralized zero-trust services and assess the role of blockchain in BES. We first build an $M/G/1$-type queuing model for BES and establish the stability conditions using matrix analytic methods. Based on the stability conditions, we identify the blockchain scalability and server capability as two critical bottlenecks of BES. We further use a tandem queuing model to describe the BES latency of the assembling and service phases. We analytically characterize the properties such as the convexity of service-phase latency with respect to traffic intensity, and highlight the BES pooling effects from traffic offloading and resource sharing. At last, we verify our conclusions through simulations and explore potential pathways for more efficient BES frameworks.**

***Index Terms*—Blockchain, Markov processes, modeling, queuing analysis.**

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I. Introduction

THE rapid development of blockchain technology has made itself a research hotspot [1]. As the backbone of cryptocurrencies, blockchain technology has shown its great value in establishing public ledgers and maintaining consensus for decentralized untrustworthy networks. This utility extends far beyond economics, with applications in diverse areas [2], [3], [4]. The defining characteristic of blockchain is its ability to facilitate various services and applications without relying on pre-established trust [5], [6]. As a result, blockchain facilitates collaboration and aggregates resources across networks with high security via a zero-trust approach [7], [8], and is recognized as a transformative force in the development of decentralized applications and networks [1], [9]. Consequently, blockchain-enabled services (BESs) are experiencing a persistent surge in interest [5], [10].

Historically, digital systems often rely on centralized intermediaries, such as banks or trusted third parties, to establish trust, but this approach often creates vulnerabilities like single points of failure and fraud risks [11], [12]. Blockchain technology addresses these issues by shifting trust from intermediaries to decentralized protocols and enabling zero-trust security where every digital action is validated through cryptographic methods and consensus mechanisms [13]. Unlike centralized models, blockchain embeds trust via immutable records, decentralized identity management [14], and automated smart contracts to enforce access policies in a decentralized approach. Also, blockchain enhances intrusion detection capabilities by providing tamper-proof logging and facilitating the threat intelligence sharing across distributed nodes [15]. These capabilities align with the zero-trust motto of "never trust, always verify," and make blockchain a foundational technology for modern cybersecurity in practical applications like secure identity verification, access control, and resource sharing [16].

Following the zero-trust principle, BES utilizes blockchain to ensure secure and reliable interactions between service providers (SPs) and users. Smart contracts in BES enforce service agreements and access policies by verifying each operation. The underlying blockchain guarantees that transactions and smart contracts between SPs and users are verified collectively by the entire network and thus eliminates the need

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for central oversight. The whole framework of BES restricts unauthorized actions and continuously monitors all interactions to ensure compliance. By adhering to this decentralized model, BES supports peer-to-peer capabilities with secure and verifiable transactions and actions that fulfill the zero-trust requirement of continuous authentication and validation [13]. BES can further integrate targeted security measures such as decentralized authentication [17], endpoint detection [15], and real-time behavioral analysis [18] to mitigate real-time risks from compromised devices or stolen credentials. Together, these capabilities allow BES to achieve reliable zero-trust services with high standards of tamper resistance and strong protection against unauthorized interventions and targeted attacks [19].

By establishing a zero-trust environment between SPs and users, smart contracts in BES lower the trust barriers that usually hinder cross-network collaboration [9], [20]. As a direct result, BES reduces the cost of establishing trust and benefits from distributed resource pooling and inter-network load balancing. Blockchain-enabled decentralized networks harness distributed computing power, storage, and bandwidth contributed by a global network of participants [5], [21]. This approach strengthens the resilience of decentralized applications and fosters innovation by enabling new forms of collaboration and sharing. Blockchain radio access network (B-RAN) is a typical example of BES for diverse wireless services between SPs and user equipments (UEs). In B-RAN, a UE can connect to a pool of accessible access points (APs) of some SPs other than its own, and access a low-traffic network when its SP is congested. Hence, from the perspective of a UE, B-RAN provides a large-scale network with more options for better connectivity at a lower cost. From an SP' s perspective, flexible coordination among hosts can effectively reduce investment in infrastructure and spectrum while drastically improving network efficiency and load balancing. Beyond B-RAN, such zero-trust services enabled by blockchain can potentially drive efficiency via inter-network collaboration in ways that traditional centralized systems cannot achieve.

However, these blockchain-based zero-trust services also come at a price. Specifically, finite block size and lengthy confirmation latency significantly limit the on-chain processing capability, which is the inherent weakness of distributed systems like blockchain. Remark that scalability is one of the major bottlenecks of BES, which directly affects the system's ability to process data efficiently and to meet the demands of increasing users and applications [6], [22], [23]. This limitation highlights the need for ongoing research to assess the efficacy of BES from a comprehensive perspective.

A. Related Work

To date, an increasing number of studies have drawn their attention to zero-trust frameworks and mechanisms based on blockchain [13], [16], [17], [18]. These studies utilized blockchain to enhance security, trust, and efficiency by automating operations, decentralizing control across the network, and eliminating the need for trusted third parties. However, despite the development of various blockchain-based zero-trust solutions, the role of blockchain and its impact on the overall system have not yet been comprehensively analyzed or assessed.

Among existing studies, some have made progress in modeling blockchain systems. The authors of [24], [25], and [26] focused on the mining process and developed queuing models to assess the scalability performance of the Bitcoin blockchain. These works inspired the subsequent studies [27], [28], [29], such as Memon et al. [29], which analyzed mining time and system throughput using a fork-join queue model. Similarly, Xu et al. [30] assessed transaction latency in the Fabric blockchain based on three independent consensus phases. Although these modeling efforts are generally solid, most of them focused only on block creation process and failed to capture the entire end-to-end blockchain-enabled service lifecycle.

Meanwhile, most studies focus only on specific system parameters [31], [32]. For example, Qi et al. [27] analyzed blockchain profit based on transaction arrival rates. Ma et al. [33] proposed a sleep blockchain queuing system and investigated the impact of sleep probability and transaction generation rate. Chakravarthy et al. [28] discussed the scalability advantages in a two-service stage blockchain system. However, these studies often overlooked the dual constraints imposed simultaneously by blockchain and service infrastructure, such as assuming infinite block size or server capacity. As a result, they failed to assess the overall system performance, including stability and scalability, for practical services and networks.

More importantly, existing literature has not quantified the specific advantages of blockchain in BES [34]. Xu et al. [10] demonstrated blockchain's potential for efficient resource management and sharing in 6G networks. Liu et al. [17] presented a blockchain-enabled information sharing scheme and highlighted security and efficiency improvements. Chen et al. [21] assessed the advantage of blockchain in enabling efficient resource offloading and reducing system costs. However, while these studies provided detailed discussions on the benefits from blockchain, their findings remain qualitative. Due to the same reason, the possible side effect of introducing blockchain has never been adequately considered. Therefore, this calls for a rigorous quantitative analysis to assess both the positive and negative impacts of blockchain on the corresponding zero-trust services.

B. Contributions

In this work, our goal is to systematically analyze and assess the role of blockchain in decentralized services and applications. By identifying both the capabilities and limitations of blockchain technology, we seek to provide a comprehensive understanding of its impacts and explore the potential pathways for an effective integration with diverse services. Through this analysis and assessment, we aim to contribute to the broader discourse on blockchain's role in shaping the future of decentralized applications and ecosystems.

The contributions of this work are summarized as follows.

- We build an $M/G/1$-type queuing model for BES and theoretically derive the stability conditions from the recurrence properties of the Markov chain by using matrix analytic methods.

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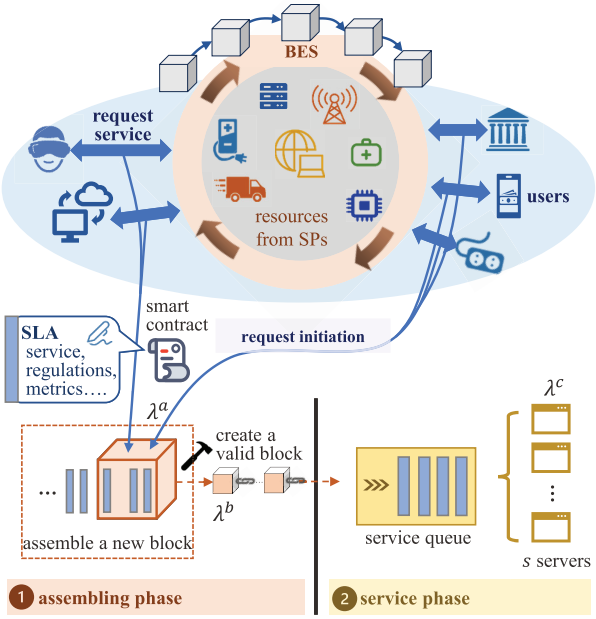


Fig. 1. Illustration of BES and its service workflow.

- Based on the stability condition, we introduce the concept of sustainable throughput regions to comprehensively assess the impact of blockchain and point out that the blockchain scalability and server capability are the two major bottlenecks of BES.
- We characterize the service phase latency of BES and prove its convexity with respect to traffic intensity. As a direct result, independent SPs with unbalanced loads can be improved by traffic offloading via blockchain.
- We further illustrate the resource sharing gain of BES, i.e., a larger shared resource pool formed by BES is more efficient than the collection of smaller non-shared SPs.
- We present a series of simulations, as well as a B-RAN case study based on real-world requests, to verify our conclusions.

The subsequent sections are organized as follows. Section II introduces BES and depicts its service workflow in the example of B-RAN. Section III and Section IV build a queuing model for BES and derive its stability conditions to reveal the major bottlenecks in BES. Section V further introduces an approximated tandem model to analyze the pooling effect led by the blockchain. Section VI presents simulation results. Finally, Section VII summarizes the paper.

II. Blockchain-Enabled Services

A. Overview

BES employs blockchain to orchestrate a suite of services across various sectors. As depicted in Fig. 1, BES enables trusted coordination among diverse SPs and ensures them to perform cooperatively and reliability through the underlying blockchain. The BES integrates a number of smart contracts to pool service resources from multiple SPs and then facilitates the secure and trusted provision, supervision, and settlement of services across subnetworks.

The BES network connects every participating SP. These SPs announce their services via smart contracts and authenticate them by their digital signatures. These smart contracts digitize the service terms, such as the shared resources, the revenue distribution, and also the exception management. As illustrated in Fig. 1, a user requesting BES signs a smart contract directly with the BES and submits it to the BES maintainers (i.e., blockchain miners). This smart contract forms a service level agreement (SLA) that specifies the expected service standards, including the services to be provided, the regulations that users and SPs must follow, the performance standards the SPs must meet, and the metrics by which service is evaluated.

Briefly, after initiating the request, the BES is established via two critical phases as follows.

1) **Assembling phase**: After receiving the smart contract, BES maintainers verify its validity, e.g., if the user has a sufficient credit balance and the service SP has enough available resources. Subsequently, BES maintainers organize a bulk of verified requests into a candidate block, perform the consensus mechanism to create a valid block (i.e., block mining), and add it on the top of the blockchain, which is further disseminated across BES network. To guarantee the tamper resistance and ensure security, the newly generated block and also the smart contracts wait until several subsequent blocks are built on top as security confirmations.
2) **Service phase**: After being confirmed by the blockchain, the smart contract enters the service queue and awaits execution. If the SP has the available resources, the service specified in the smart contract is delivered. Otherwise, the smart contract lines up in the service queue until an idle resource is available.

The above two steps of BES enable zero trust security by applying decentralized verification, immutable certification, and continuous monitoring into service delivery [35]. In the assembling phase, SPs work together to verify smart contracts through the consensus mechanism that prevents any single SP from altering or approving transactions alone. This phase checks whether the user' s credit balance and the SP' s resources meet the requirements, and only verified requests proceed further. The consensus mechanism then organizes verified requests into blocks, which are added to the blockchain and shared across the network. Through this process, the decentralized consensus and the blockchain' s immutable ledger protect against tampering and provide a transparent, tamper-resistant audit trail. In the service phase, once the blockchain confirms the smart contract, SPs execute the service request if the required resources remain available. This approach of assembling and then verifying each step through multiple SPs reduces the risk of unauthorized actions or breaches. BES can further implement decentralized identity management and dynamic authentication measures such as risk-based access control [36], continuous multi-factor authentication [37], and biometric verification [38] to enhance overall security. Through these protocol-based rules and cryptographic safeguards, BES maintains continuous verification to uphold

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TABLE I
SYMBOL DEFINITIONS

Symbols	Explanations
K	Number of SPs in BES
λ_k^a	Request arrival rate of SP k
$\lambda^a = \sum_{k=1}^{K} \lambda_k^a$	Sum request arrival rate in BES
λ^b	Block generation rate
λ^c	Service rate
m	Maximum block size limit
s_k	Number of servers of SP k
$s = \sum_{k=1}^{K} s_k$	Total number of servers in BES
$\zeta = \frac{\lambda^a}{m\lambda^b}$	Assembling intensity
$\rho = \frac{\lambda^a}{s\lambda^c}$	Service intensity
$T_k^a = \frac{1}{\lambda_k^a}$	Average inter-arrival time
$T^b = \frac{1}{\lambda^b}$	Average block time
$T^c = \frac{1}{\lambda^c}$	Average service time
N	Number of required confirmations

secure service coordination under zero-trust principles, without relying on centralized trust [35].

B. System Model

Mathematically, we can describe BES consisting of K SPs via the following model. The requests are assumed as a Poisson process, with rate λ_k^a for SP k. Because of the additive property of Poisson process, the total request arrival rate of BES is $\lambda^a = \sum_{k=1}^{K} \lambda_k^a$. Denote s_k as the number of available servers in SP k, i.e., SP k can serve at most s_k users simultaneously. Therefore, BES has $s = \sum_{k=1}^{K} s_k$ servers, which can be viewed as the service capability. Every request's service time is assumed to be mutually independent and exponential with mean $1/\lambda^c$. In this sense, λ^c can be viewed as the service rate, i.e., how fast the server is.

BES is compatible with multiple consensus protocols. In this work, we focus on the proof of work (PoW) protocol, which is the most commonly used in practice [5]. The block generation process can be modeled as a Poisson process with rate λ^b, that is, λ^b blocks are expected to be generated per unit time [39]. Our modeling work is suitable and can be easily extended to most proof-based consensus mechanism. For security considerations, every request requires N blocks to confirm before service. For example, $N = 6$ is used in applications such as cryptocurrencies as a rule of thumb [26]. In this study, we specifically consider the one-confirmation scenario where $N = 1$. (As investigated in [39], the number of confirmations N balances the trade-off between security and latency. The analysis for the multiple confirmation setup can be extended according to the method in [39].) Furthermore, we denote the maximum block size limit as m, i.e., at most m requests can be put into one block. According to the definition of the above symbols, we define $\zeta = \frac{\lambda^a}{m\lambda^b}$ and $\rho = \frac{\lambda^a}{s\lambda^c}$ as the assembling intensity and service intensity, respectively. All the symbols in the paper are summarized in Table I.

C. B-RAN: An Example of BES

Now we would like to introduce B-RAN as an example of BES for wireless access services [5]. B-RAN incorporates blockchain-based mechanisms for secure and trustworthy wireless services, including access, authentication, authorization, resource sharing, data interaction, and information supervision. It supports dynamic collaboration among mobile network operators (MNOs) and covers various wireless scenarios like cellular communication, Internet of Things, and mobile edge computing.

B-RAN aggregates multiple MNOs, serving as SPs, into a larger combined network. This integration creates a wireless resource pool that includes spectrum, storage, computation, and infrastructure. Users can directly sign a smart contract, endorsed by B-RAN SPs, to request wireless services. Once B-RAN maintainers validate this smart contract via the consensus mechanism, a suitable member SP is assigned to handle the service request. SPs can also redistribute traffic more efficiently, particularly during peak times by offloading excessive traffic within a single subnetwork, thus preventing failures and maintaining consistent access.

In general, B-RAN can integrate uncoordinated APs within radio access networks (RANs) to improve reliability and also expand coverage. This integration allows entities to commodify their assets on a growing, ubiquitous network, while capitalizing on the positive network effects. The above insight is supported by the experiments of our prior works [5], [6]. Hence, a collaborative network of SPs in B-RAN is expected to be more efficient than operating independently. However, such a finding still lacks a rigorous proof as the theoretical foundation.

III. BES MARKOV MODEL

As a typical BES, B-RAN is analytically characterized by a queuing model in [39], which is named B-RAN Markov model by [40]. Thereby, B-RAN Markov model is also effective for most BESs. The model selects the state space $(I(t), J(t))$ to utilize the Markovian and time-homogeneous properties, where $I(t)$ and $J(t)$ represent the number of pending requests and confirmed ones (in the one-confirmation case) at time t, respectively. In this approach, the two-dimensional stochastic process $\{(I(t), J(t)), t \geq 0\}$ can be used to describe the queuing in BES with the state probability $w_{i,j}(t) = \Pr\{(I(t), J(t)) = (i, j)\}$. Define $w_{i,j} = \lim_{t\to\infty} w_{i,j}(t)$ as the stationary probability and thus $\{w_{i,j}\}$ is the steady-state distribution.

As shown in Section II-A, the stochastic process $(I(t), J(t))$ is driven by three types of events that are request arrivals (in the preparing phase), block generations (in the assembling phase), and service completions (in the service phase). The state switches from (i, j) to $(i+1, j)$ if a new service request arrives, whereas it moves to $(i, j-1)$ if a service is complete. Note that when a new block is generated, only up to m pending requests can be assembled into the candidate block and subsequently added to the blockchain. Consequently, the state transitions from (i, j) to $(i - \min(i, m), j + \min(i, m))$. Based on the derivations in

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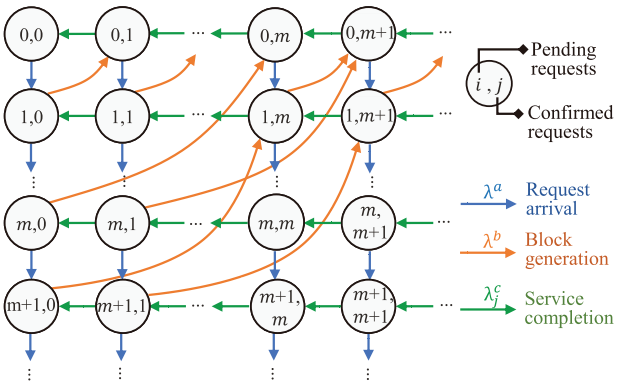


Fig. 2. The state transition diagram for BES.

Section IV of [39], $(I(t), J(t))$ is proven to be a continuous time-homogeneous Markov process, i.e., the future of the process is independent of the past given only its present state, and the transition probabilities are independent of the starting time. Given that the inter-request time, inter-block time, and service time obey exponential distribution with rate λ^a, λ^b, and λ^c, respectively, the equilibrium equations of the queuing model are given by:

$$w_{i,j}\left(\lambda^a+\lambda_j^c\right)=\sum_{k=1}^{m} w_{k,j-k}\lambda^b+w_{i,j+1}\lambda_{j+1}^c,$$
$$i=0, j=0,1,\ldots, \tag{1}$$
$$w_{i,j}\left(\lambda^a+\lambda^b+\lambda_j^c\right)=w_{i-1,j}\lambda^a+w_{i,j+1}\lambda_{j+1}^c,$$
$$i=1,2,\ldots, j=0,\ldots,m-1, \tag{2}$$
$$w_{i,j}\left(\lambda^a+\lambda^b+\lambda_j^c\right)=w_{i-1,j}\lambda^a+w_{i+m,j-m}\lambda^b+w_{i,j+1}\lambda_{j+1}^c,$$
$$i=1,2,\ldots, j=m,m+1,\ldots, \tag{3}$$

where $\lambda_j^c=\min(j,s)\lambda^c$ represents the service completion rate of BES. Using this model, we can obtain the steady-state distribution $\{w_{i,j}\}$ of BES by solving (1)–(3) along with the sum probability condition $\sum_{i,j} w_{i,j}=1$.

IV. Stability Conditions

A. M/G/1-Type Queuing Model

Stability is a fundamental and essential requirement for queuing systems. The stability of BES is crucial to ensure that the steady-state distribution derived in Section III is both achievable and reliable. A stable BES will converge to equilibrium regardless of initial conditions. It ensures that transient states stabilize over time and prevents unbounded growth or oscillations. In this section, we will derive the stability conditions to provide a comprehensive picture of the impact of blockchain on BES.

First, as depicted in Fig. 2, we re-arrange the two-dimensional state space $(I(t), J(t))$ described in Section III by an alphabetical order captured by the stationary probability vector

$$\mathbf{w}=\left[\mathbf{w}_0\middle|\mathbf{w}_1\middle|\mathbf{w}_2\middle|\cdots\right],$$
$$\mathbf{w}_j=[w_{0,j},w_{1,j},w_{2,j},\ldots], j=0,1,\ldots$$

The balance equations (1)–(3) can then be reorganized into a vector form. The coefficients in these equations are subsequently structured into the transition rate matrix $\mathbf{Q}$, which can be written as a block-matrix form:

$$\mathbf{Q}=\begin{bmatrix}\mathbf{A}_{1,0} & \mathbf{A}_2 & \mathbf{A}_3 & \ldots & \mathbf{A}_{m+1} & & \\ \mathbf{A}_{0,1} & \mathbf{A}_{1,1} & \mathbf{A}_2 & \ldots & \mathbf{A}_m & \mathbf{A}_{m+1} & \\ & \mathbf{A}_{0,2} & \mathbf{A}_{1,2} & \ldots & \mathbf{A}_{m-1} & \mathbf{A}_m & \ddots \\ & & \mathbf{A}_{0,3} & \ldots & \mathbf{A}_{m-2} & \mathbf{A}_{m-1} & \ddots \\ & & & \ddots & \ldots & \ddots & \ddots \end{bmatrix}, \tag{4}$$

where

$$\mathbf{A}_{0,j}=\lambda_j^c\mathbf{I}, \quad \mathbf{A}_k=\lambda^b\mathbf{E}_{k,1}, 2\le k<m+1,$$
$$\mathbf{A}_{m+1}=\sum_{i=1}^{\infty}\lambda^b\mathbf{E}_{m+i,i},$$
$$\mathbf{A}_{1,j}=\begin{bmatrix}-\lambda^a-\lambda_j^c & \lambda^a & & \\ & -\lambda^a-\lambda^b-\lambda_j^c & \lambda^a & \\ & & -\lambda^a-\lambda^b-\lambda_j^c & \ddots \\ & & & \ddots \end{bmatrix}.$$

$\mathbf{I}$ and $\mathbf{O}$ represent the identity matrix and the zero matrix with proper dimensions, respectively. $\mathbf{E}_{i,j}$ is the matrix whose (i,j)-th entry is one and all other entries are zeros.

By properly arranging the state space, the transition rate matrix $\mathbf{Q}$ has a structure similar to that of an $M/G/1$ queue, where all entries below and to the left of the subdiagonal are zero. (In Kendall notation, an $M/G/1$ queue refers to a system where arrivals follow a Markovian (M) process and service times obey a general (G) distribution, with a single (1) server.) That is why our model is named as an $M/G/1$ queue. The $M/G/1$-type model encodes the state transition relation into blocks to construct a more compact structure. We denote the level j consisting of the states as $j=\{(i,j)\,|i=0,1,\ldots\}$. The structure of $\mathbf{Q}$ indicates the transitions from level j to level $j+\ell-1$ for $j\ge 0$ and $\ell\ge 0$ are governed by entries of $\mathbf{A}_\ell$. The maximum level that can be crossed in a single transition is limited by the maximum block size m. Thanks to the properties of $\mathbf{Q}$, we can derive the stability conditions from the bottom.

B. Stability Analysis

In this subsection, we identify the stability conditions of the BES using the $M/G/1$-type model. We first define the stability mathematically in the following.

Definition 1: A continuous time Markov chain possesses stability if the stationary distribution $\mathbf{w}$ which is uniquely determined by the equilibrium equation $\mathbf{wQ}=\mathbf{0}^T$ exists, where $\mathbf{Q}$ is the transition rate matrix.

Stability implies that the Markov chain possesses a well-defined long-term behavior. According to Definition 1, there exists an invariant regime described by the measure $\mathbf{w}$, where if the chain initiates within this regime, it persists indefinitely. If the chain starts in any other state, it will, with overwhelming probability, converge to this invariant regime, with $\mathbf{w}$ as its limiting distribution. Essentially, the queue will consistently

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stabilize at a viable level as described by $\mathbf{w}$. In Lemma 1, we further clarify the meaning of stability based on Theorem 3.7 in [41].

Lemma 1: Consider a Markov chain satisfying the irreducibility and positive recurrence, then the vector $\mathbf{w}$ *determined by the equation* $\mathbf{wQ} = \mathbf{0}^T$ *forms a stationary distribution.*

From Lemma 1, we can see that the stability is equivalent to two properties, i.e., irreducibility and positive recurrence. Apparently, the expression of (4) shows that the Markov chain $(I(t), J(t))$ is irreducible. Therefore, we focus the positive recurrent condition. To address it, we use the matrix analytic methods from Neuts in [42] and [43], which is summarized in the following Lemma 2.

Lemma 2: If the matrix $\mathbf{A} = \sum_{k=0}^{\infty} \mathbf{A}_k$ *is a stochastic matrix, the irreducible Markov process of* $M/G/1$ *type is positive recurrent if and only if*

$$\max\{\boldsymbol{\theta}\mathbf{A}_{0,j}\mathbf{1}\} > \sum_{k=2}^{\infty}(k-1)\,\boldsymbol{\theta}\mathbf{A}_k\mathbf{1}, \tag{5}$$

where $\boldsymbol{\theta} = (\theta_0, \theta_1, \ldots)$ *is given by the unique solution to the system of linear equations:* $\boldsymbol{\theta}\mathbf{A} = \mathbf{0}^T$ *and* $\boldsymbol{\theta}\mathbf{1} = 1$*, and* $\mathbf{1}$ *is a column vector of ones with proper dimension.*

Now we give the stable conditions of BES in the following theorem.

Theorem 1: BES system possesses stability if and only if

$$\lambda^a < m\lambda^b, \text{i.e., blockchain scalability,} \tag{6}$$

$$\lambda^a < s\,\lambda^c, \text{i.e., server capability.} \tag{7}$$

Proof: Please see Appendix-A.■

In Theorem 1, $m\lambda^b$ specifies the maximum rate at which requests can be recorded by the blockchain, whereas $s\lambda^c$ characterizes the maximum processing capability of BES. Hence, Theorem 1 indicates that the request arrival rate λ^a must be lower than both of these rates to ensure a stable BES. Intuitively, blockchain scalability and server capability are the two main bottlenecks of BES.

If the stability condition in (6) is not met, the blockchain becomes the primary bottleneck. This inefficiency in handling requests causes them to accumulate infinitely during the assembling phase and leads to server under-utilization in BES. This degradation may further discourage SPs from participating in BES, as they might prefer serving users directly if the blockchain cannot manage incoming requests effectively. Conversely, if (7) is not satisfied, server capacity becomes the limiting factor. This scenario overburdens every server from all participating SPs in the BES system and causes a backlog in the service queue, even by introducing collaboration among SPs.

In both failure scenarios, user experience deteriorates rapidly due to delays of handling requests in an unstable BES system. This will ultimately reduce both SPs' and users' willingness to engage with BES. Such a decline could collapse the trust and cooperation between users, SPs, and the BES system. Therefore, it is crucial to ensure the stability of the BES system by carefully considering both blockchain scalability and server capability.

C. Sustainable Throughput Region

In the following, we would like to show the impact of blockchain on the service capability of BES. For the SP k, the sustainable throughput of BES is defined as $s_k\lambda^c$, which represents the maximum service rate the SP k can provide. Then we compare the sustainable throughput of BES with the case where SPs all operate independently. For K independent SPs, the total sustainable throughput region is simply a K-dimensional cube with the hypervolume

$$V_I^K = (\lambda^c)^K \prod_{k=1}^{K} s_k.$$

Meanwhile, BES can integrate SPs by aggregating their resources and balancing traffics. Therefore, the sustainable throughput region of BES is a K-dimensional simplex with the hypervolume

$$V_S^K = (\lambda^c)^K \frac{s^K}{K!}.$$

Their sustainable throughput regions are illustrated in Fig. 4 in Section VI. Within this sustainable throughput region, the SPs can operate stably and continuously provide BES.

Now, we use the ratio of sustainable throughput to show the advantages of BES, which is given by

$$\frac{V_S^K}{V_I^K} = \frac{s^K}{K!\prod_{k=1}^{K} s_k} = \frac{\left(\sum_{k=1}^{K} s_k\right)^K}{K!\prod_{k=1}^{K} s_k}. \tag{8}$$

Utilizing the arithmetic mean-geometric mean (AM-GM) inequality, we further have

$$\frac{V_S^K}{V_I^K} \geq \frac{K^K}{K!}. \tag{9}$$

The equality holds if and only if $s_1 = s_2 = \cdots = s_K$. From (9), the ratio is always higher than one, which means BES could unite SPs into a shared larger network with an aggregated and stronger service capability. Also, we can compare the K-SP case with the $(K+1)$-SP one:

$$\frac{V_S^{K+1}V_I^K}{V_S^K V_I^{K+1}} = \frac{\left(\sum_{k=1}^{K} s_k + s_{K+1}\right)^{K+1}}{(K+1)\,s_{K+1}\left(\sum_{k=1}^{K} s_k\right)^K} \geq 1. \tag{10}$$

From (10), one can observe that the ratio gradually rises as the number of SPs increases. (Note that, the ratio tends to the infinity if some s_k is close to zero.) As more SPs join the union to provide BES, the sustainable throughput of the entire network also grows more rapidly. This shows that a larger-scale network of SPs, or simply more SPs joining BES instead of operating independently, will make the positive network effect more evident.

Nevertheless, the scalability of blockchain also degrades the sustainable throughput of BES. In the assembling phase, the number of requests arriving during the mean block time must be less than the maximum block size. Otherwise, BES maintainers cannot include all requests in the blockchain, leading to a growing backlog of requests waiting to be assembled into a block. Hence, as shown in (6), the request arrival rate must satisfy $\lambda^a < m\lambda^b$, where m represents the scalability of

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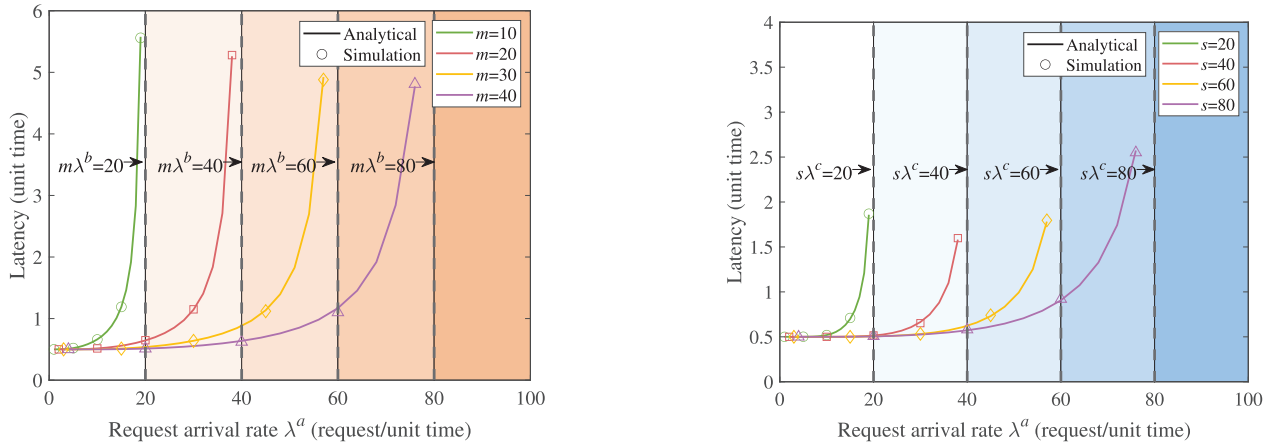


Fig. 3. Stability conditions of BES under varying λ^a. (a) Different m (λ^c=1 request/unit time,s=100 request). (b) Different s (λ^b=2 block/unit time,m=50 block).

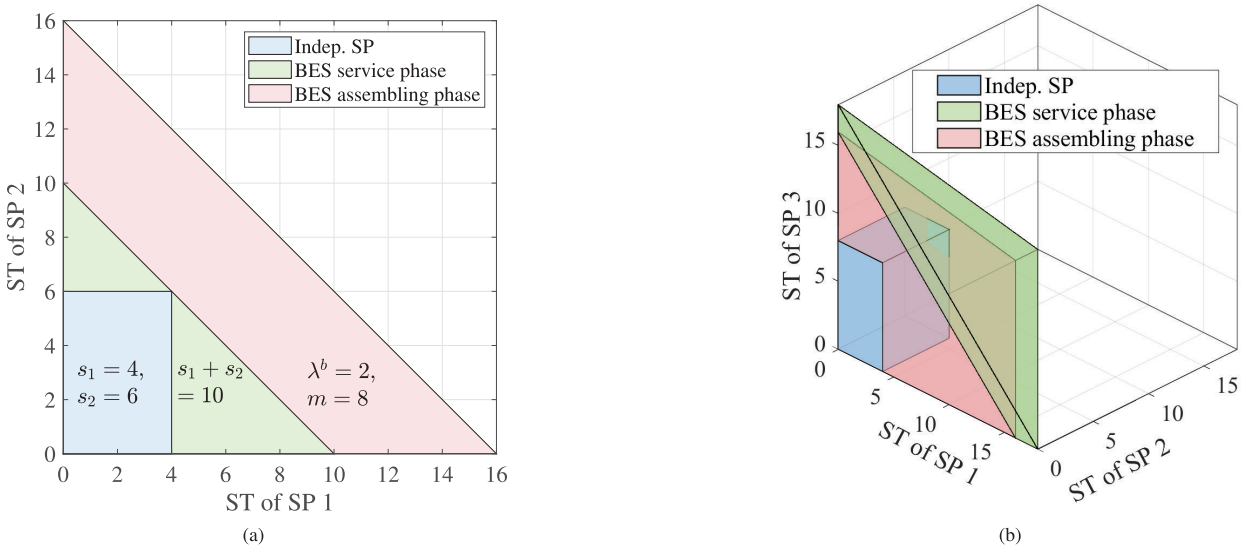


Fig. 4. The sustainable throughput (ST) region of BES and independent SPs. (a) 2 SPs. (b) 3 SPs.

blockchain. Constrained by the blockchain, the hypervolume of the sustainable throughput region of BES becomes

$$V_B^K = \left(\lambda^b\right)^K \frac{(mK)^K}{K!},$$

where V_B^K represents the sustainable throughput limited by the blockchain scalability. While V_S^K indicates a greater overall service capability of BES compared to V_I^K, V_B^K indicates that the blockchain scalability also limits the throughput of BES. Therefore, V_S^K and V_B^K imply the constraints on the sustainable throughput region of BES caused by service capability and blockchain scalability, which are consistent with the stability conditions in Theorem 1.

V. Pooling Effect Assessment

A. Tandem Model

Section IV-A derives the stability conditions for a general BES. Even though a B-RAN Markov model and its variant $M/G/1$-type model have been established, it remains challenging to quantitatively assess the role of blockchain with some closed-form expressions. Therefore, in this subsection, we attempt to develop a tandem model with proper simplification to analytically characterize the pooling effect of BES.

Consider that the birth-death process with a tridiagonal transition rate matrix is easier to analyze. Inspired by this, we construct an approximated transition rate matrix $\mathbf{Q}^A$ with a block-tridiagonal form. From the structure of $\mathbf{Q}$ in (4), one can see that the block-tridiagonal form is precluded by the matrices $\{\mathbf{A}_k\}_{k=2}^{m+1}$. Therefore, we integrate the matrices $\{\mathbf{A}_k\}_{k=2}^{m+1}$ into $\mathbf{A}_{1,j}$ for each row and let the entries to the right of the diagonal be zero. To avoid the Markov chain being a pure death process, we further introduce extra transition relations to model the birth process, that is, every level has independent single births continuously at rate λ^a. In this approach, the new transition rate matrix $\mathbf{Q}^A$ is written as

$$\mathbf{Q}^A = \begin{bmatrix} \tilde{\mathbf{A}}_{1,0} & \mathbf{B} & & \\ \mathbf{A}_{0,1} & \tilde{\mathbf{A}}_{1,1} & \mathbf{B} & \\ & \mathbf{A}_{0,2} & \tilde{\mathbf{A}}_{1,2} & \ddots \\ & & \ddots & \ddots \end{bmatrix}, \tag{11}$$

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where

$$\tilde{\mathbf{A}}_{1,j} = \mathbf{A}_{1,j} + \sum_{k=2}^{m+1} \mathbf{A}_k - \lambda^a \mathbf{I}$$
$$= \begin{bmatrix} -2\lambda^a - \lambda_j^c & \lambda^a & & & \\ \lambda^b & -2\lambda^a - \lambda^b - \lambda_j^c & \lambda^a & & \\ \vdots & & -2\lambda^a - \lambda^b - \lambda_j^c & \ddots & \\ \lambda^b & & & \ddots & \\ & \lambda^b & & & \\ & & & \ddots & \end{bmatrix},$$
$$\mathbf{B} = \lambda^a \mathbf{I}.$$

By carefully observing the state transition relations of $\mathbf{Q}^A$, we identify the following characteristics. If a new BES request arrives, the state switches from (i, j) to either $(i+1, j)$ or $(i, j+1)$ with rate λ^a, i.e., the request arrival may have an impact on both the assembling phase and service phase. If a new block is generated, the state moves from (i, j) to $(i - \min(i, m), j)$ with rate λ^b, i.e., block generation can only affect the assembling phase. If a BES completes (obviously $j \geq 1$ at this instant), the state shifts from (i, j) to $(i, j-1)$ with rate λ_j^c, which is the same as $\mathbf{Q}$. The main difference between $\mathbf{Q}$ and $\mathbf{Q}^A$ comes from the transitions of block generation. In the case of $\mathbf{Q}$, at most m requests can be assembled into a candidate block and further be added into the blockchain so that the state changes from (i, j) to $(i - \min(i, m), j + \min(i, m))$. With regard to $\mathbf{Q}^A$, similarly, up to m pending requests can be packed, resulting in a decrease in the number of pending requests from i to $i - \min(i, m)$. However, these requests are included into the blockchain individually and the number of confirmed requests increases one by one.

Essentially, $I(t)$ is an $M(\lambda^a)/M^{X(t)}(\lambda^b)/1$ queue, where a newly generated block at the epoch t has the size $X(t) = \min(I(t^-), m)$ and t^- represents the epoch just before the time t. Meanwhile, $J(t)$ forms an $M(\lambda^a)/M(\lambda^c)/s$ queue. Therefore, the approximated transition rate matrix $\mathbf{Q}^A$ precisely characterizes the two queues in tandem. That is how the name of tandem model comes. Mathematically, we further find that the established tandem model is equivalent to the single arrival model in [40], which has been shown as a great approximation. As shown in Section VI, the accurate service latency firmly agrees with the approximated ones, which shows the effectiveness of the tandem model. From the tandem model, the service latency $\mathbb{E}[T_Q]$ consists of the assembling phase latency $\mathbb{E}[T^A]$ and the service phase latency $\mathbb{E}[T_Q^S]$:

$$\mathbb{E}[T_Q] = \mathbb{E}[T^A] + \mathbb{E}[T_Q^S].$$

In the following, we would like to evaluate BES performance mathematically based on the tandem model and assess the benefits from pooling.

B. Traffic Offloading Gain

Traffic offloading is an effective approach to alleviating network congestion under unbalanced traffics. BES supports traffic offloading by promoting collaboration among a group of participating SPs. Thereby, it can effectively manage traffic loads, particularly during peak periods and sudden surges, by redistributing resources to maintain stability and reduce service costs. For example, when SP 1 faces high traffic volumes, it can initiate a smart contract that outlines the transfer conditions and billing for offloaded traffic. If SP 2 agrees, it confirms the smart contract and allows SP 1 to redirect excess traffic to SP 2, thus preventing network overload and potential collapse.

Mathematically, we define $\mathbb{E}[T_Q^S; \rho]$ as the service phase latency of SP, given the service intensity ρ. The theoretical benefit of this approach is summarized in Theorem 2.

Theorem 2: Given λ^b, λ^c, m, and s, as the service intensity ρ, or equivalently λ^a, increases, the service phase latency $\mathbb{E}[T_Q^S; \rho]$ is convex in respect to ρ.

Proof: Please see Appendix-B.■

The convexity in Theorem 2 suggests that the traffic offloading strategy can effectively reduce the service phase latency, $\mathbb{E}[T_Q^S; \rho]$, in BES where the traffics of subnetworks are unbalanced. We summarize the traffic offloading gain of BES in the following corollary.

Corollary 1: BES yields a shorter (at least not longer) service phase latency $\mathbb{E}[T_Q^S]$ compared to an independent service process without traffic offloading. The 2-SP case can be mathematically expressed as

$$\underbrace{\mathbb{E}\left[T_Q^S; \frac{1}{2}(\rho_1 + \rho_2)\right]}_{\text{Average service latency}} \leq \underbrace{\frac{\rho_1 \mathbb{E}[T_Q^S; \rho_1]}{\rho_1 + \rho_2}}_{\text{Service latency of SP 1}} + \underbrace{\frac{\rho_2 \mathbb{E}[T_Q^S; \rho_2]}{\rho_1 + \rho_2}}_{\text{Service latency of SP 2}}, \tag{12}$$

where the equality holds only if $\rho_1 = \rho_2$.

Proof: Consider two SPs with the same number of servers s. By the convexity of $\mathbb{E}[T_Q^S; \rho]$ with respect to ρ, Jensen' s inequality implies

$$\mathbb{E}\left[T_Q^S; \frac{1}{2}(\rho_1 + \rho_2)\right] \leq \frac{\mathbb{E}[T_Q^S; \rho_1] + \mathbb{E}[T_Q^S; \rho_2]}{2} \leq \frac{\rho_1 \mathbb{E}[T_Q^S; \rho_1]}{\rho_1 + \rho_2} + \frac{\rho_2 \mathbb{E}[T_Q^S; \rho_2]}{\rho_1 + \rho_2},$$

where the right-hand side represents the average service phase latency of the independent SPs. The first inequality follows from the convexity of $\mathbb{E}[T_Q^S]$ with respect to λ^a. To prove the second inequality, we assume $\rho_1 > \rho_2$ without loss of generality, which implies $\mathbb{E}[T_Q^S; \rho_1] > \mathbb{E}[T_Q^S; \rho_2]$. The weighted average assigns a larger weight to the larger term $\mathbb{E}[T_Q^S; \rho_1]$ and thus exceeds the arithmetic mean, proving the second inequality. The equality in (12) holds only if $\rho_1 = \rho_2$, or equivalently, $\lambda_1^a = \lambda_2^a$, i.e., the traffic intensity of each SP is identical. The above proof can be extended to multiple SPs with arbitrary numbers of servers.■

Theorem 2 and Corollary 1 together illustrate how BES can manage uneven or fluctuating traffic loads across various SPs. These results indicate that SPs facing unbalanced loads can benefit from traffic offloading through BES. Specifically, during peak traffic periods, SPs can deploy traffic offloading smart contracts to distribute loads more evenly and fairly and

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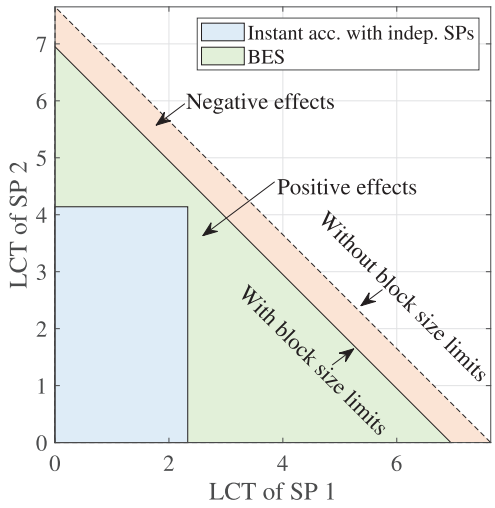


Fig. 5. The LCT region of BES and two independent SPs.

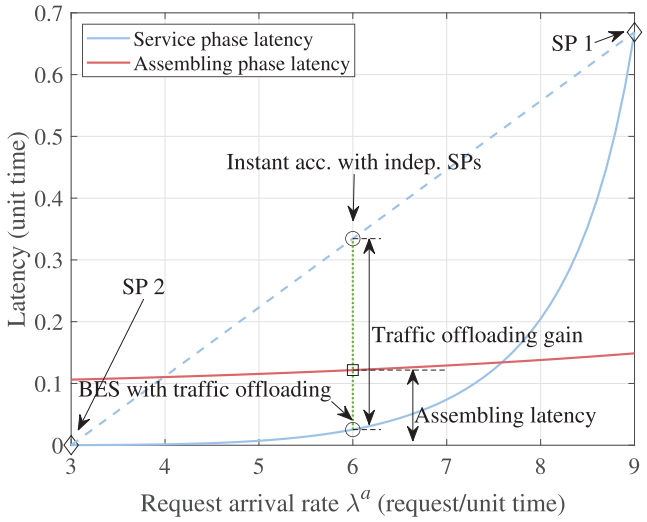


Fig. 6. The latency of BES and two independent SPs with and without traffic offloading.

thus provide higher service qualities. These results highlight the pivotal role that blockchain plays in facilitating effective collaboration among SPs and showcase the pooling effects from traffic offloading. The convexity of the delay function $\mathbb{E}\left[T_Q^S; \rho_i\right]$ provides a theoretical foundation for BES performance analysis, which is visualized in Fig. 6 in Section VI.

C. Network Sharing Gain

Section V-B demonstrates the traffic offloading gain in BES. Now, we would like to show that BES also exhibits sharing gain even with balanced traffics. The network sharing gain indicates that a larger shared network formed by multiple SPs, such as BES, is more efficient than the collection of smaller non-shared ones. That is the essential reason why BES can benefit from the pooling of SPs and resource aggregation.

Specifically, we would like to show in the following theorem that, for a fixed traffic intensity ρ, the BES with a larger number of servers s, which means that it integrates resources from many SPs and also has to handle their combined traffics, λ^a, exhibits greater processing capability and higher efficiency. We explicitly present the network sharing gain of BES as follows.

Theorem 3: Given λ^b, λ^c, and m, with a fixed service intensity ρ, the service phase latency $\mathbb{E}\left[T_Q^S\right]$ decreases with the increase of the server number s.

Proof: Please see Appendix-C.■

Theorem 3 indicates that, given service intensity ρ, the service phase latency $\mathbb{E}\left[T_Q^S\right]$ of BES drops as the number of servers s increases. The resources are pooled and shared across SPs, resulting in a converged network with greater service capability. Note that the stronger processing capability and shorter service latency are achieved despite the BES processing the larger combined traffics from different SPs.

Theorem 3 further suggests that the capability of a BES network consisting of multiple SPs exceeds that of the sum of smaller, non-shared networks. As more SPs join and collaborate, the efficiency of the entire shared network increases, even without advanced allocation algorithms. Moreover, since BES operates in a permissionless manner, the enhanced service quality resulting from network sharing gains attracts more users. This, in turn, motivates more SPs to participate. Such positive feedback between users and SPs continuously expands BES and thus enhances the value of the entire network.

VI. Simulation Results

In this section, using wireless services as an example, we illustrate the impact of blockchain in BES based on our lab-built B-RAN prototype [39], [44]. In the experiments, we set the number of block confirmations to $N = 1$.[1] The service rate is normalized to be $\lambda^c = 1$ and thus the latency in this study is presented relative to the average service time, i.e., the unit time. We show the impact of the key network parameters, such as the request arrival rate λ^a, the assembling intensity ζ, and the traffic intensity ρ, to illustrate the pros and cons of blockchain. We compare the performance of BES with that of a model based on independent SPs, also known as "instant access strategy" [45]. In the benchmark, SPs operate independently, and clients obtain services from their subscribed SP. All the experiments in this work are based on a single workstation equipped with an Intel I7-13700K processor, 32 gigabytes RAM, and 2 terabytes storage.

We first verify the stability conditions of BES in Fig. 3, with $\lambda^b = 2$ fixed. In Fig. 3(a), the number of servers is relatively large ($s = 100$), which ensures that the stability condition of the service phase $\lambda^a < s\lambda^c$ holds. This allows us to observe the impact of blockchain scalability on system stability. As λ^a approaches $m\lambda^b$, latency becomes lengthy significantly. By increasing m, the stability region expands, as long as the stability condition $\lambda^a < m\lambda^b$ is maintained. In Fig. 3(b), the maximum block size is relatively large ($m = 50$) to guarantee $\lambda^a < m\lambda^b$. When λ^a approaches $s\lambda^c$, latency sharply increases. This highlights the stability condition $\lambda^a < s\lambda^c$ for the service phase. Intuitively, Fig. 3(a) denotes the stability conditions limited by the blockchain

[1]For simplicity, the setup with $N > 1$ is not explicitly included in the simulations. It is straightforward that the BES latency with N confirmations would increase by an additional $(\lambda^b)^{-1}(N-1)$ compared to the case with a single confirmation [39].

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scalability $m\lambda^b$, while Fig. 3(b) reveals the constraint imposed by the network processing capability $s\lambda^c$. The stability of BES requires both conditions to be satisfied simultaneously. Fig. 3 also illustrates the resilience of BES under extreme service intensities. As m or s increases, the BES system can accommodate higher incoming traffic without latency spikes or service interruptions. In addition, Fig. 3 suggests that operating BES under extreme service intensities is not recommended, as it brings the system closer to instability and leads to an unbounded and significant increase in service latency.

To illustrate the performance of BES compared to independently operating SPs, we analyze the sustainable throughput regions shown in Fig. 4. In Fig. 4(a), we consider a BES with two SPs ($\lambda^b = 2, m = 8, s_1 = 4, s_2 = 6$). For independent SPs, the sustainable throughput region is represented by a rectangle with an area of $\lambda^c s_1 \times \lambda^c s_2$. In contrast, by sharing resources through blockchain, the number of servers in BES increases to $s_1 + s_2 = 10$. This expands the sustainable throughput region, considering only the service phase, into a larger triangle with an area of $\left(\lambda^c (s_1 + s_2)\right)^2 / 2$, which completely covers the rectangular area. The sustainable throughput region, considering the assembling phase, also forms a triangle with an area of $\left(m\lambda^b\right)^2 / 2$. This area encompasses both regions of the BES service phase and independent SPs, thus not limiting BES performance. In Fig. 4(b), we add a third SP ($s_3 = 8$) without changing the assembling capacity. The sustainable throughput region of the service phase in BES forms a triangular pyramid, which fully contains the cube representing the case of independent SPs. However, the overall sustainable throughput region of BES is restricted by the assembling phase. This indicates the scalability issues with blockchain. Furthermore, the sustainable throughput regions of the service phase in both cases are tangent. It shows that the benefits of blockchain have yet to be fully demonstrated.

Therefore, to better illustrate the throughput performance of BES, we introduce the concept of latency-constrained throughput (LCT). This is defined as the maximum request arrival rate, λ^a, that satisfies the latency-violation probability constraint $\Pr\{T_Q \geq t_{\max}\} \leq \varepsilon$, with $t_{\max}$ as the maximum latency bound and ε as the target violation probability. In Fig. 5, we set $t_{\max} = 1$ and $\varepsilon = 0.05$. After resource sharing and traffic offloading, the LCT region of BES fully encompasses that of the instant access benchmark with independent SPs. This indicates a significant throughput improvement and highlights the positive impact of blockchain. Note that the block size narrows the region, which can be viewed as a negative impact of blockchain. Fig. 5 comprehensively visualizes the impact of blockchain on BES performance.

Next, we demonstrate the capabilities of BES in uniting SPs and balancing traffic loads. Fig. 6 compares latency before and after traffic offloading in the BES scenario with the instant access benchmark with independent SPs. In Fig. 6, two SPs each have 10 servers ($s_1 = s_2 = 10$) with arrival rates $\lambda_1^a = 9$ and $\lambda_2^a = 3$, respectively. The block generation rate and maximum block size of BES are set to $\lambda^b = 10$ and $m = 2$. As shown in Fig. 6, the service phase latency is a convex function of λ^a, which strongly supports Theorem 2. BES facilitates cooperation among SPs by offloading excess traffic from SP 1 to SP 2 via blockchain and thereby balancing traffic load. Due to the convex nature of the service phase latency, the latency after traffic offloading in BES is significantly lower than before, as depicted by the green dotted line, which can be regarded as the traffic offloading gain. As shown in Fig. 6, the assemble phase adds an extra latency, which is also the negative impact of blockchain. In general, Fig. 6 illustrates that BES can effectively reduce the overall service latency, consistent with the theoretical analysis in Section V-B. As the network scale expands, the traffic offloading gain will also increase. This demonstrates the pooling effect of BES.

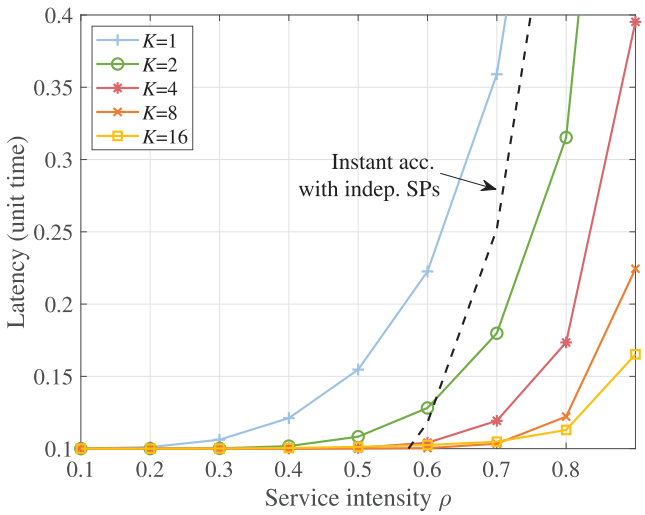


Fig. 7. The latency of BES and independent SPs under varying K and ρ.

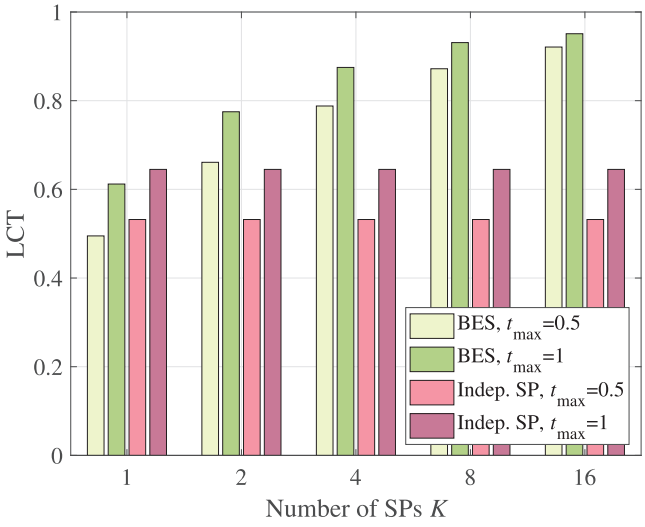


Fig. 8. The LCT of BES and independent SPs under different $t_{\max}$.

Furthermore, we illustrate the benefits of BES from resource sharing and the system resilience against heavy traffic loads in Fig. 7 and Fig. 8. The BES configuration parameters are $\lambda^b = 5$, $m = 40$, $s_k = 10$, with λ_k^a varying with service intensity ρ. The arrival rate and number of servers increase linearly with the number of SPs, K, i.e., $\lambda^a = K\lambda_k^a$ and $s = Ks_k$. Fig. 7 compares BES service latency under different numbers of SPs. Increasing K significantly reduces latency, especially in high-intensity scenarios, which highlights the resource sharing gain of BES. The instant access benchmark

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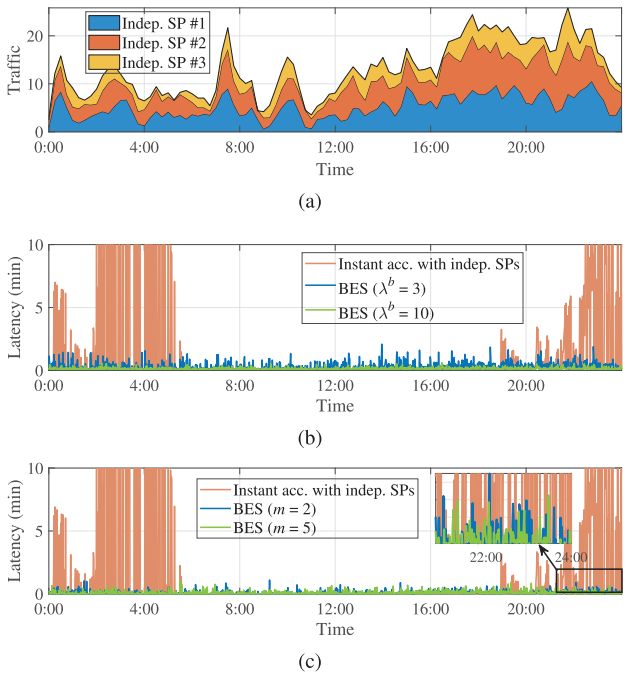


Fig. 9. Performance comparison of BES and instant access benchmark with independent SPs over time in practical scenarios. (a) Real traffic. (b) Latency under different λ^b. (c) Latency under different m.

with independent SPs, however, suffers high latency under heavy loads due to the lack of inter-SP resource sharing. Fig. 7 shows the BES benefits from larger shared networks and exhibits better performance than independent SPs. It can be seen that BES handles high-traffic bursts better and offers greater robustness than independent SPs.

Fig. 8 shows the network sharing gain of BES through LCT, normalized by the number of servers s. Initially, BES throughput may be lower than that of independent SPs due to the blockchain scalability issue and assembling latency. However, the resource sharing gain becomes prominent with more SPs. Overall, Fig. 6, Fig. 7, and Fig. 8 demonstrate traffic offloading and network sharing gains of BES. As more SPs collaborate, the pooling effect of BES becomes more significant.

Finally, Fig. 9 compares the service performance of BES and instant access with independent SPs using a real-world dataset from the Google analytics platform [46]. This dataset can reflect realistic user traffic patterns, including arrival times and service durations, and capture dynamic fluctuations in request flows, which no longer obey the Poisson assumptions. Specifically, we use customer visit records on 21 July 2017. Fig. 9(a) shows traffic distribution across three independent SPs and highlights significant temporal variations with peak traffic from 11 am to 1 pm. Fig. 9(b) and Fig. 9(c) further illustrate the comparative performance of BES and instant access with independent SPs. The benchmark exhibits more pronounced fluctuations, especially during high traffic bursts, while BES shows smoother performance overall. Under high traffic intensity, BES significantly reduces latency due to inter-SP offloading and resource sharing. Fig. 9 demonstrates the strong adaptability of BES to high-intensity burst traffics. In low traffic scenarios, both cases have low latencies, but BES experiences additional delays from achieving network-wide consensus, implying the limitations of blockchain. Moreover, larger λ^b and m can further reduce latency and mitigate traffic fluctuation impacts.

VII. Conclusion

In this paper, we provide a thorough analysis of the role of blockchain in BES. We first establish an $M/G/1$-type queuing model to determine the stability conditions of BES and use the sustainable throughput region to view the full picture. From the stability conditions, we highlight that the blockchain scalability and server capability are the two major bottlenecks of BES. We then build a tandem queuing model, including the assembling and service stages, to evaluate the impact of blockchain on pooling resources. On the one hand, our analysis reveals that BES can enhance overall performance via load balancing, i.e., the blockchain-enabled traffic offloading gain. On the other hand, we also prove that, as the network scales, the service-phase latency becomes shorter, even under balanced traffic for every SP, which is summarized as resource sharing gain. In conclusion, the BES pooling effect derives from the combined benefits of traffic offloading and resource sharing, which are verified by a group of simulations with a real-data case study.

Appendix

A. Proof of Theorem 1

Proof: For the queuing model in Section III, we have

$$\mathbf{A} = \mathbf{A}_{0,j} + \mathbf{A}_{1,j} + \sum_{k=2}^{m+1} \mathbf{A}_k = \begin{bmatrix} -\lambda^a & \lambda^a & & & & \\ \lambda^b & -\lambda^a-\lambda^b & \lambda^a & & & \\ \vdots & & \ddots & \ddots & & \\ \lambda^b & & & -\lambda^a-\lambda^b & \lambda^a & \\ & \lambda^b & & & -\lambda^a-\lambda^b & \ddots \\ & & \ddots & & & \ddots \end{bmatrix},$$

and $\boldsymbol{\theta} = (\theta_0, \theta_1, \theta_2, \ldots)$ is also the unique solution of $\mathbf{A}$. Obviously, the matrix $\mathbf{A}$ is a stochastic matrix with each row summing to zero. The matrix form of linear equations $\boldsymbol{\theta}\mathbf{A} = \mathbf{0}^T$ can be expressed as

$$-\lambda^a\theta_0 + \lambda^b \sum_{i=1}^{m} \theta_i = 0, \tag{13}$$

$$-\left(\lambda^a + \lambda^b\right)\theta_i + \lambda^a\theta_{i-1} + \lambda^b\theta_{i+m} = 0, i \geq 1. \tag{14}$$

According to (14), the solution $\boldsymbol{\theta}$ exhibits the geometric property and yields

$$\theta_0 = 1 - r, \tag{15}$$

$$\theta_i = \theta_0 r^i, i = 1, 2, \ldots, \tag{16}$$

where r is a constant. By substituting (15) and (16) into (13) and (14), we have

$$r^{m+1} - r - \lambda^a/\lambda^b\,(r-1) = 0,$$

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$$-\left(\lambda^a+\lambda^b\right)r^i+\lambda^a r^{i-1}+\lambda^b r^{i+m}=0. \tag{17}$$

Then we can rearrange (17) into

$$\sum_{j=1}^{m} r^j=\lambda^a/\lambda^b. \tag{18}$$

To ensure the existence of the solution $\boldsymbol{\theta}$, the constant r must satisfy $r<1$. It is worthwhile to note that $r<1$ if and only if $\lambda^a/\lambda^b<m$, which indicates

$$\lambda^a<m\lambda^b.$$

According to Lemma 2, the left-hand side of (5) can be written as

$$\max\left\{\boldsymbol{\theta}\mathbf{A}_{0,j}\mathbf{1}\right\}=\max\left\{\lambda_j^c\right\}=s\lambda^c,$$

and the right-hand side can be expressed as

$$\begin{aligned}\sum_{k=2}^{m+1}(k-1)\,\boldsymbol{\theta}\mathbf{A}_k\mathbf{1}&=\lambda^b\left(m-\theta_0\sum_{i=0}^{m-1}(m-i)\,r^i\right)\\&=\lambda^b\left(m-\sum_{i=0}^{m}\left(1-r^i\right)\right).\\&=\lambda^b\sum_{i=1}^{m}r^i=\lambda^a.\end{aligned}$$

Hence, we finally obtain:

$$\lambda^a<s\lambda^c.$$

Due to the irreducibility of the Markov chain, (6) and (7) ensure positive recurrence and thereby guarantee stability. In conclusion, (6) and (7) are the stability conditions of BES.■

B. Proof of Theorem 2

Proof: Let π_i denote the steady-state probability of i requests being in the service phase of BES (including those waiting for and currently receiving service), and N_Q represent the number of requests queuing for the service. Based on the basic results of queues [47], we have $\pi_0=\left(\sum_{i=0}^{s-1}\frac{(s\rho)^i}{i!}+\frac{(s\rho)^s}{s!(1-\rho)}\right)^{-1}$, $\pi_s=\frac{(s\rho)^s}{s!}\pi_0$, and $\mathbb{E}\left[N_Q\right]=\frac{\rho}{(1-\rho)^2}\pi_s$. The derivative of π_0 with respect to ρ is

$$\begin{aligned}\frac{d\pi_0}{d\rho}&=-\pi_0^2\left(s\sum_{i=0}^{s-1}\frac{(s\rho)^i}{i!}+s\frac{(s\rho)^s}{s!\,(1-\rho)}+\frac{(s\rho)^s}{s!\,(1-\rho)^2}\right)\\&=-\pi_0\left(s+\frac{\mathbb{E}\left[N_Q\right]}{\rho}\right).\end{aligned} \tag{19}$$

Then the derivative of π_s is

$$\frac{d\pi_s}{d\rho}=\frac{s^s}{s!}\left(s\rho^{s-1}\pi_0+\rho^s\frac{d\pi_0}{d\rho}\right)=\frac{s-s\rho-\mathbb{E}\left[N_Q\right]}{\rho}\pi_s. \tag{20}$$

The derivative of $\mathbb{E}\left[N_Q\right]$ is given by

$$\begin{aligned}\frac{d\mathbb{E}\left[N_Q\right]}{d\rho}&=\frac{\left(\pi_s+\rho\frac{d\pi_s}{d\rho}\right)(1-\rho)^2+2\rho\pi_s\,(1-\rho)}{(1-\rho)^4}\\&=\mathbb{E}\left[N_Q\right]\left(\frac{1+s-s\rho-\mathbb{E}\left[N_Q\right]}{\rho}+\frac{2}{1-\rho}\right).\end{aligned} \tag{21}$$

We further take the derivative of $\frac{d\mathbb{E}[N_Q]}{d\rho}$:

$$\begin{aligned}\frac{d^2\mathbb{E}\left[N_Q\right]}{d\rho^2}&=\frac{\left(\frac{d\mathbb{E}[N_Q]}{d\rho}\right)^2}{\mathbb{E}\left[N_Q\right]}+\mathbb{E}\left[N_Q\right]\\&\quad\times\left(-\frac{\frac{d\mathbb{E}[N_Q]}{d\rho}}{\rho}+\frac{\mathbb{E}\left[N_Q\right]-s-1}{\rho^2}+\frac{2}{(1-\rho)^2}\right),\end{aligned} \tag{22}$$

where $\mathbb{E}\left[N_Q\right]$ and $\frac{d\mathbb{E}[N_Q]}{d\rho}$ are not constrained. Therefore, we replace all $\mathbb{E}\left[N_Q\right]$ and $\frac{d\mathbb{E}[N_Q]}{d\rho}$ in (21) and (22) with the probability P_Q that an arriving request has to queue (i.e., the probability that all servers are busy). P_Q is easily bounded as the expected number of busy servers is $s\rho$, which means

$$\sum_{i=0}^{s-1}i\pi_i+sP_Q=s\rho,$$

and therefore $P_Q\le\rho$. Now, we can rewrite (21) and (22) by $\mathbb{E}\left[N_Q\right]=\frac{\rho}{1-\rho}P_Q$ as:

$$\begin{aligned}\frac{d\mathbb{E}\left[N_Q\right]}{d\rho}&=P_Q\left(s+\frac{1+\rho-\rho P_Q}{(1-\rho)^2}\right),\\\frac{d^2\mathbb{E}\left[N_Q\right]}{d\rho^2}&=P_Q\frac{1-\rho}{\rho}\left(s+\frac{1+\rho-\rho P_Q}{(1-\rho)^2}\right)^2+\frac{\rho}{1-\rho}P_Q\\&\quad\times\left(-\frac{sP_Q}{\rho}-\frac{(1+\rho-\rho P_Q)\,P_Q}{\rho(1-\rho)^2}+\frac{P_Q}{\rho(1-\rho)}\right.\\&\quad\left.-\frac{s+1}{\rho^2}+\frac{2}{(1-\rho)^2}\right).\end{aligned}$$

Hence, we have (23), as shown at the bottom of the page. Since

$$\begin{aligned}&\frac{(1-\rho P_Q)^2}{\rho\,(1-\rho)^3}\\&\quad=\frac{\left(1-\rho^2+\rho^2-\rho P_Q\right)^2}{\rho\,(1-\rho)^3}\\&\quad=\frac{(1+\rho)^2}{\rho\,(1-\rho)}+\frac{2\,(1+\rho)\,(\rho-P_Q)}{(1-\rho)^2}+\frac{\rho\,(\rho-P_Q)^2}{(1-\rho)^3},\end{aligned}$$

we have

$$\begin{aligned}\frac{1}{P_Q}\frac{d^2\mathbb{E}\left[N_Q\right]}{d\rho^2}&=\frac{1}{(1-\rho)^3}\left(\rho\,(\rho-P_Q)^2+2-P_Q\right.\\&\quad\left.+3\rho\,(1-P_Q)+\rho P_Q^2\right)\end{aligned}$$

$$\frac{1}{P_Q}\frac{d^2\mathbb{E}\left[N_Q\right]}{d\rho^2}=\frac{(1-\rho P_Q)^2}{\rho\,(1-\rho)^3}+\frac{2-P_Q+3\rho\,(1-P_Q)+\rho P_Q^2}{(1-\rho)^3}+\frac{s\left(\frac{1}{\rho}-\frac{1}{s\rho}+2-3P_Q\right)}{1-\rho}+\frac{(1-\rho)\,s^2}{\rho}+\frac{P_Q}{(1-\rho)^2}. \tag{23}$$

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$$+\frac{s}{1-\rho}\left(\frac{1}{\rho}-P_Q+2\left(1-P_Q\right)\right)+\frac{2+\rho}{1-\rho}+\frac{2\left(1+\rho\right)\left(\rho-P_Q\right)+P_Q}{\left(1-\rho\right)^2}+\frac{\left(1-\rho\right)s^2}{\rho}.$$

Since $1-\rho, \rho-P_Q, 1-P_Q$, and $\frac{1}{\rho}-P_Q$ are all non-negative, we thus have

$$\frac{d^2\mathbb{E}\left[N_Q\right]}{d\rho^2}\geq 0, \quad \forall 0<\rho<1. \tag{24}$$

As shown in (24), $\mathbb{E}\left[N_Q\right]$ is convex in ρ, which completes the proof that $\mathbb{E}\left[T_Q^S;\rho\right]$ is also convex with respect to ρ.■

C. Proof of Theorem 3

Proof: The service phase latency $\mathbb{E}\left[T_Q^S\right] = \frac{\rho(s\rho)^s}{\lambda^a(1-\rho)^2 s!}\left(\sum_{i=0}^{s-1}\frac{(s\rho)^i}{i!}+\frac{(s\rho)^s}{s!(1-\rho)}\right)^{-1}$. Define $g(s) \triangleq \sum_{i=0}^{s-1}\frac{s!(s\rho)^{i-s}}{i!}$. Now we prove that $g(s)$ is monotonically increasing in s. Let $j=i-1$ and we have

$$g\left(s+1\right)=\sum_{j=0}^{s-1}\frac{s!\left(s\rho\right)^{j-s}}{j!\left(1+\frac{1}{s}\right)^{s-j}}\frac{s+1}{j+1}+\frac{(s+1)!}{\left(\left(s+1\right)\rho\right)^{s+1}}.$$

To prove $g\left(s+1\right) > g\left(s\right)$, we would like to show $\frac{1}{\left(1+\frac{1}{s}\right)^{s-i}}\cdot\frac{s+1}{i+1}\geq 1$ for any $i=0,1,\ldots,s-1$. Define

$$h\left(i\right)\triangleq \ln\left(\frac{1}{\left(1+\frac{1}{s}\right)^{s-i}}\frac{s+1}{i+1}\right),$$

and its first-order derivative is

$$h'\left(i\right)=\ln\left(1+\frac{1}{s}\right)-\frac{1}{i+1}.$$

From $\ln\left(1+\frac{1}{s}\right)\leq\frac{1}{s}$ for $s>0$, $\ln\left(1+\frac{1}{s}\right)\leq\frac{1}{s}\leq\frac{1}{i+1}$ always holds for $0\leq i\leq s-1$. Hence, we have $h'\left(i\right)\leq 0$ such that $h\left(i\right)\geq h\left(s-1\right)=0$, i.e., $\frac{1}{\left(1+\frac{1}{s}\right)^{s-i}}\frac{s+1}{i+1}\geq 1$ for $i=0,1,\ldots,s-1$. Now we conclude that

$$\begin{aligned} g\left(s+1\right)&\geq\sum_{i=0}^{s-1}\frac{s!\left(s\rho\right)^{i-s}}{i!}+\frac{(s+1)!}{\left(\left(s+1\right)\rho\right)^{s+1}}\\ &>\sum_{i=0}^{s-1}\frac{s!\left(s\rho\right)^{i-s}}{i!}=g\left(s\right).\end{aligned}$$

Therefore, as s increases, $g\left(s\right)$ also increases such that $\mathbb{E}\left[T_Q^S\right]$ decreases, which proves Theorem 3.■

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